

Algebra: Chapter 0 - Self Study

Grant Talbert

Winter Break, 2024/2025



College of Arts and Sciences
Department of Mathematics and Statistics
CAS MA 741

Abstract

This text contains my notes on Algebra: Chapter 0, by Paolo Aluffi. I read this book independently, starting over winter break 2024. My notes largely paraphrase the text, and include few examples. I have also completed every exercise in an external document. I also completed every problem set from [this Algebra Course at Johns Hopkins University](#). I read chapters 1, 2, and 3 over the course of winter break, and read chapters 4-9 over the spring semester. I did not read them in order; I read chapter 6, 8, and 9 before most of chapters 4 and 5.

Contents

1	Linear Algebra, Reprise	2
1.1	Preliminaries, Reprise	2
1.1.1	Functors	2
1.1.2	Examples of Functors	3
1.1.3	When are Two Categories ‘Equivalent’?	4
1.1.4	Limits and Colimits	5
1.1.5	Comparing Functors	7
1.2	Tensor Products and the Tor Functor	11
1.2.1	Bilinear Maps and the Definition of Tensor Product	11
1.2.2	Adjunction with Hom and Explicit Computations	13
1.2.3	Exactness Properties of Tensor; Flatness	15
1.2.4	The Tor Functors	16

Chapter 1

Linear Algebra, Reprise

1.1 Preliminaries, Reprise

1.1.1 Functors

Throughout this book, we have seen natural ways to go from one category to another, such as $\mathbf{Ring}$ to $\mathbf{Ab}$, or $\mathbf{Grp}$ to $\mathbf{Set}$, or $\mathbf{Fld}_k$ to $k\text{-}\mathbf{Vect}$. However, we have not needed to formalize this notion until now. We will do this with functors.

Definition 1.1: Functor

Let $\mathbf{C}, \mathbf{D}$ be categories. A *covariant* functor $\mathcal{F} : \mathbf{C} \rightarrow \mathbf{D}$ is an assignment of an object $\mathcal{F}(A) \in \mathbf{Obj}(\mathbf{D})$ for all $A \in \mathbf{Obj}(\mathbf{C})$, and of a function, also denoted $\mathcal{F}$:

$$\mathcal{F} : \mathbf{Hom}_{\mathbf{C}}(A, B) \rightarrow \mathbf{Hom}_{\mathbf{D}}(\mathcal{F}(A), \mathcal{F}(B))$$

for all objects $A, B \in \mathbf{Obj}(\mathbf{C})$, such that

- Identity maps are preserved:

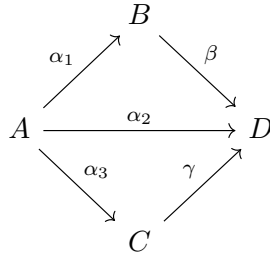
$$\mathcal{F}(1_A) = 1_{\mathcal{F}(A)}.$$

- Composition is preserved:

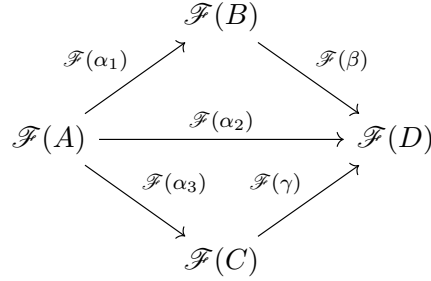
$$\mathcal{F}(\beta \circ \alpha) = \mathcal{F}(\beta) \circ \mathcal{F}(\alpha).$$

A *contravariant* functor $\mathcal{G} : \mathbf{C} \rightarrow \mathbf{D}$ is a covariant functor $\mathbf{C}^{op} \rightarrow \mathbf{D}$.

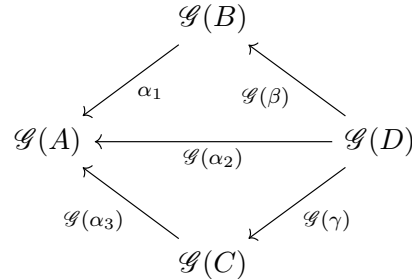
The fact that contravariant functors still preserve morphisms means $\mathcal{G}(\alpha \circ \beta) = \mathcal{G}(\beta) \circ \mathcal{G}(\alpha)$. This all means that diagrams are preserved: a diagram



is sent to a diagram



and a contravariant functor sends it to a diagram



In either case, commutative diagrams are sent to commutative diagrams.

Contravariant functors are also often called *presheaves*. A presheaf of sets on a category $\mathbf{C}$ is given by a contravariant functor $\mathbf{C} \rightarrow \mathbf{Set}$. You can have similar structures as well; for example, a presheaf of abelian groups on a category $\mathbf{C}$ is a contravariant functor $\mathbf{C} \rightarrow \mathbf{Ab}$.

If the Hom-sets of a category carry more structure, we may ask that functors preserve this structure. For example, we have seen that $\text{Hom}_R(-, -)$ is an abelian group, so we may ask that a functor $\mathcal{F} : R\text{-Mod} \rightarrow S\text{-Mod}$ be an abelian group homomorphism

$$\text{Hom}_R(A, B) \rightarrow \text{Hom}_S(\mathcal{F}(A), \mathcal{F}(B))$$

for all R -modules A, B . A functor preserving this specific structure in this way is called *additive*.

1.1.2 Examples of Functors

There are many examples of functors that “forget” the structure of a category; for example, $\mathbf{Ring} \rightarrow \mathbf{Set}$ in the natural way. Such functors are called *forgetful functors*.

If R is a ring, then we denote by R^* the *group of units* in R . Every ring homomorphism $R \rightarrow S$ also induces a group homomorphism $R^* \rightarrow S^*$, and this assignment is compatible with compositions (we say such an assignment is functorial). Thus, there is a covariant functor $\mathbf{Ring} \rightarrow \mathbf{Grp}$ defined in this way.

We have previously defined the spectrum $\text{Spec}(R)$ of a ring R as the *set* of prime ideals of R . If R, S are rings and $\phi : R \rightarrow S$ is a ring homomorphism, then the inverse image $\phi^{-1}(\mathfrak{p})$ of a prime ideal is prime. Thus, ϕ induces the set function

$$\phi^* : \text{Spec}(S) \rightarrow \text{Spec}(R).$$

This is clearly a functorial assignment, so we can view the spectrum as a contravariant functor $\text{Spec} : \mathbf{Ring} \rightarrow \mathbf{Set}$.

If S is a set, recall that there exists a category $\hat{S}$ whose objects are subsets of S , and whose morphisms $U \rightarrow V$ are inclusions $U \subseteq V$. A *presheaf of sets* on S is a contravariant functor $\hat{S} \rightarrow \mathbf{Set}$. A more interesting variant of this occurs when considering a topological space T . One can then define a category in a similar way, where the objects are open sets in T , leading to the notion of a presheaf on a topological space. Standard constructions such as continuous complex-valued functions on open sets of T can be viewed

as a presheaf of abelian groups on T . Concrete examples such as this often satisfy additional hypotheses making them *sheaves*, but this is a topic for a book on algebraic topology.

An important class of functors is defined as follows. Let $\mathbf{C}$ be a category, and let X be an object of $\mathbf{C}$. Then the maps

$$A \mapsto \text{Hom}_{\mathbf{C}}(X, A), \quad A \mapsto \text{Hom}_{\mathbf{C}}(A, X)$$

define covariant and contravariant functors, respectively. These are denoted $\text{Hom}_{\mathbf{C}}(X, _)$ and $\text{Hom}_{\mathbf{C}}(_, X)$, respectively. For example, if we have the diagram

$$\begin{array}{ccccc} A & \xrightarrow{\alpha} & B & \xrightarrow{\beta} & C \\ & \searrow \xi_A & \downarrow \xi_B & \swarrow \xi_C & \\ & & X & & \end{array}$$

then by requiring the diagram to commute, we clearly have a notion of a function $\text{Hom}_{\mathbf{C}}(B, X) \rightarrow \text{Hom}_{\mathbf{C}}(A, X)$ for every choice of α by mapping $\xi_B \mapsto \xi_A$. The triangle on the right shows clearly that this assignment is contravariant. The functor $\text{Hom}_{\mathbf{C}}(_, X)$ is often denoted h_X if the category is understood.

Note that $h_X : \mathbf{C} \rightarrow \mathbf{Set}$, so we have found that there exists an assignment for *every* object of $\mathbf{C}$ a presheaf of sets on $\mathbf{C}$. The question then becomes *when is a presheaf $\mathcal{F}$ of sets on $\mathbf{C}$ in fact equal to h_X for some object X in $\mathbf{C}$?* Additionally, *when can the object X be reconstructed from the presheaf h_X ?* For example, if p is the final object of $\mathbf{Top}$, that is, p is the one-point topological space, then we have that $h_X(p)$ is nothing but X , when the single function $X \rightarrow p$ is viewed as a set. This is why geometric contexts often call $h_X(A)$ the set of A -points of X , and h_X is the functor of points of X .

1.1.3 When are Two Categories ‘Equivalent’?

Every notion insofar of isomorphism has boiled down to “structure-preserving bijection.” It is not unexpected then that the notion of equivalence for two categories is very similar, although it is not exactly the same. Such a notion would be suprisingly restrictive. Every universal construction we have encountered is only defined up to isomorphism, so requiring solutions to match precisely between categories would be difficult. We instead want to attempt and match isomorphism classes, rather than individual objects.

Definition 1.2: Faithful/Full

Let $\mathbf{C}, \mathbf{D}$ be categories. A covariant functor $\mathcal{F} : \mathbf{C} \rightarrow \mathbf{D}$ is *faithful* if for all objects A, B of $\mathbf{C}$, the induced function

$$\text{Hom}_{\mathbf{C}}(A, B) \rightarrow \text{Hom}_{\mathbf{D}}(\mathcal{F}(A), \mathcal{F}(B))$$

is injective. A functor is *full* if the function is *surjective*. A fully faithful functor is one where the function is bijective.

Morphisms are the most important information in a category, and fully faithful functors preserve morphisms. It is also easily shown that they preserve isomorphism classes, and thus a fully faithful functor can in general identify a category $\mathbf{C}$ with a full subcategory of $\mathbf{D}$. From here, we can construct our definition of an equivalence of categories.

Definition 1.3: Equivalence of Categories

A covariant functor $\mathcal{F} : \mathbf{C} \rightarrow \mathbf{D}$ is an *equivalence of categories* if it is fully faithful and *essentially surjective*; that is, for all $Y \in \text{Obj}(\mathbf{D})$, there exists $X \in \text{Obj}(\mathbf{C})$ such that $\mathcal{F}(X) = Y$. We say $\mathbf{C}$ and $\mathbf{D}$ are *equivalent*.

It is not immediate, but equivalence of categories is in fact an equivalence relation. Additionally, an equivalence of categories $\mathcal{F} : \mathbf{C} \rightarrow \mathbf{D}$ has an inverse in a weaker sense; that is, there exists a functor $g : \mathbf{D} \rightarrow \mathbf{C}$ such that $\mathcal{F}g$ and $g\mathcal{F}$ are isomorphic in a natural way to the identity map. This fact will not be used in this book very extensively, so we do not prove it.

For example, consider the category where objects are nonnegative integers, and let $\text{Hom}(m, n)$ be the set of $n \times m$ matrices with entries in a field k , and with composition defined by product of matrices. The resulting category is equivalent in a natural way to the subcategory of $k\text{-Vect}$ consisting of finite-dimensional vector spaces. The functor is clearly fully faithful, and it is essentially surjective because all vector spaces are free, and are thus classified by dimensionality.

1.1.4 Limits and Colimits

The universal properties we have encountered throughout the book are all examples of the categorical notion of a limit.

Definition 1.4: Limit

Let $\mathcal{F} : \mathbf{I} \rightarrow \mathbf{C}$ be a covariant functor, where one thinks of $\mathbf{I}$ as a category of indices. The *limit* of $\mathcal{F}$, if it exists, is an object L of $\mathbf{C}$, endowed with morphisms $\lambda_I : L \rightarrow \mathcal{F}(I)$ for all objects I of $\mathbf{I}$, satisfying

- For any morphism $\alpha : I \rightarrow J$ in $\mathbf{I}$, then $\lambda_J = \mathcal{F}(\alpha) \circ \lambda_I$:

$$\begin{array}{ccc} & L & \\ \lambda_I \swarrow & & \searrow \lambda_J \\ I & \xrightarrow{\mathcal{F}(\alpha)} & J \end{array}$$

- L is final with respect to this property; that is, if M is another object together with morphisms μ_I satisfying this property, then there exists a unique morphism $L \rightarrow M$ such that all relevant diagrams commute. In particular,

$$\begin{array}{ccc} & L, & \\ \lambda_I \swarrow & \downarrow \phi & \searrow \lambda_J \\ & M & \\ \mu_I \swarrow & & \searrow \mu_J \\ I & \xrightarrow{\mathcal{F}(\alpha)} & J \end{array}$$

must commute.

Remark

Any such M together with morphisms μ_I is called a *cone* for $\mathcal{F}$; a limit is then a final object in the category of cones of $\mathcal{F}$.

The limit, denoted $\varprojlim \mathcal{F}$, is unique up to isomorphism if it exists, as are all constructions characterized by a universal property. The left pointing arrow is a reminder that L comes “before” all objects indexed by $\mathcal{F}$, and projects outwards along the morphisms λ_I . This limit is also called a projective limit.

For example, consider the universal property of products. Let $\mathbf{I}$ be the discrete category consisting of two objects $\{1, 2\}$ with only the identity morphisms, and let $\mathcal{A} : \mathbf{I} \rightarrow \mathbf{C}$ be a functor such that $\mathcal{A}(1) = A_1$

and $\mathcal{A}(2) = A_2$, for A_1, A_2 objects in $\mathbf{C}$. Then since there exists no such α as described in the diagrams above, we have that for all objects X of $\mathbf{C}$, the following commutes:

$$\begin{array}{ccc}
 & & A_1 \\
 & \nearrow f & \nearrow \pi_{A_1} \\
 X & \xrightarrow{\phi} \varprojlim \mathcal{A} & \\
 & \searrow g & \searrow \pi_{A_2} \\
 & & A_2
 \end{array}$$

That is, $\varprojlim \mathcal{A}$ is the *product* $A_1 \times A_2$, provided the limit exists. We can generalize this notion, and define a product of a possibly infinite collection of objects as the limit of a functor from the corresponding discrete category, provided it exists.

The limits are more exciting if we give $\mathbf{I}$ more structure. Consider the category where the entirety of the morphisms are described by the diagram

$$\hookrightarrow 1 \begin{array}{c} \xrightarrow{\alpha} \\ \xleftarrow{\beta} \end{array} 2 \hookrightarrow$$

A functor $\mathcal{K} : \mathbf{I} \rightarrow \mathbf{C}$ then amounts to a choice of two objects of $\mathbf{C}$ and a choice of two parallel morphisms between them. Limits of such functors are called *equalizers*. For example, given the category $R\text{-Mod}$, we can let $\phi : A_2 \rightarrow A_1$ be a homomorphism and choose $\mathcal{K}(\alpha) = \phi$ and $\mathcal{K}(\beta) = 0$. Then it will be shown that the limit of this functor, that is, the equalizer of ϕ and the 0 homomorphism, is simply $\ker \phi$.

Another typical example is when $\mathbf{I}$ consists of an ordered set. For example, let the objects of $\mathbf{I}$ be all positive integers $\mathbb{Z}^+$, and let there exist a unique morphism $i \rightarrow j$ if and only if $i \leq j$. A functor $\mathcal{F} : \mathbf{I} \rightarrow \mathbf{C}$ then amounts to a choice of objects A_i and morphisms $\phi_{ji} : A_i \rightarrow A_j$ for all $i \in \mathbb{Z}^+$, with the requirement that $\phi_{ii} = 1_{A_i}$ and $\phi_{kj}\phi_{ji} = \phi_{ki}$. This is the same as choosing a diagram

$$\cdots \xrightarrow{\phi_{45}} A_4 \xrightarrow{\phi_{34}} A_3 \xrightarrow{\phi_{23}} A_2 \xrightarrow{\phi_{12}} A_1$$

in $\mathbf{C}$, since all morphisms can be recovered from choices of morphisms $A_{i+1} \rightarrow A_i$. A limit $\varprojlim \mathcal{F}$, which may also be denoted

$$\varprojlim_i A_i$$

when the morphisms are clear from context, is an object A with morphism $\phi_i : A \rightarrow A_i$ such that the diagram

$$\begin{array}{ccccccc}
 & & A & & & & \\
 & \nearrow \dots & \nearrow \phi_4 & \downarrow \phi_3 & \nearrow \phi_2 & \nearrow \phi_1 & \\
 \cdots & \xleftarrow{\phi_{45}} A_4 & \xrightarrow{\phi_{34}} A_3 & \xrightarrow{\phi_{23}} A_2 & \xrightarrow{\phi_{12}} A_1 & &
 \end{array}$$

commutes, and such that all other objects satisfying this property factor uniquely through A .

Claim. The aforementioned limit $\varprojlim_i A_i$ exists in $R\text{-Mod}$.

This proof is done in the exercises.

The dual notion to a limit of a functor $\mathcal{F} : \mathbf{I} \rightarrow \mathbf{C}$ is a *colimit*.

Definition 1.5: Colimit

A *colimit* of a functor $\mathcal{F} : \mathbf{I} \rightarrow \mathbf{C}$ is an object C of $\mathbf{C}$ endowed with morphisms $\gamma_I : \mathcal{F}(I) \rightarrow C$ for all objects I of $\mathbf{I}$, such that

- For any morphism $\alpha : I \rightarrow J$ in $\mathbf{I}$, we have $\gamma_I = \gamma_J \circ \mathcal{F}(\alpha)$. We then say C is a *co-cone* of $\mathcal{F}$.
- C is *initial* with respect to this requirement. That is C is initial with respect to co-cones of $\mathcal{F}$.

The colimit of $\mathcal{F}$, if it exists, is denoted $\varinjlim \mathcal{F}$, and is also called the injective limit.

We have already encountered many examples of colimits, similarly to limits. For example, the coproduct of A_1, A_2 in some category is simply the colimit of the functor from the discrete category with two objects. Cokernels are colimits, just as kernels are limits.

We can once again consider an example with the totally ordered set $\mathbf{I}$, but with arrows reversed

$$1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 4 \longrightarrow \dots$$

A functor then consists of a choice of elements and morphisms

$$A_1 \xrightarrow{\psi_{12}} A_2 \xrightarrow{\psi_{23}} A_3 \xrightarrow{\psi_{34}} A_4 \xrightarrow{\psi_{45}} \dots$$

A colimit $\varinjlim_i A_i$ is then an object A such that there exist morphisms ψ_i such that the diagram

$$\begin{array}{ccccccc} A_1 & \xrightarrow{\psi_{12}} & A_2 & \xrightarrow{\psi_{23}} & A_3 & \xrightarrow{\psi_{34}} & A_4 & \xrightarrow{\psi_{45}} & \dots \\ & \searrow \psi_1 & & \searrow \psi_2 & \downarrow \psi_3 & & \searrow \psi_4 & \searrow \psi_5 & \\ & & & & A & & & & \end{array}$$

commutes, and A is initial with respect to this.

An example of this explicit construction is in the category **Set**, where ψ_{ji} is injective. We are then considering (up to isomorphism) a nested sequence

$$A_1 \subseteq A_2 \subseteq A_3 \subseteq A_4 \subseteq \dots$$

The direct limit is the infinite union $\bigcup_i A_i$. These constructions can be extended to posets without much difficulty, as will be done in the exercises. Colimits are used in many standard constructions, such as germs of functions, and stalks of sheaves.

1.1.5 Comparing Functors

Functors are introduced as morphisms between categories. Now we look at morphisms between functors as a way to compare two functors together.

Definition 1.6: Natural Transformation

Let $\mathbf{C}, \mathbf{D}$ be categories, and let $\mathcal{F}, \mathcal{G} : \mathbf{C} \rightarrow \mathbf{D}$ be covariant functors. A natural transformation $\nu : \mathcal{F} \Rightarrow \mathcal{G}$ is given by a collection of morphisms $\nu_X : \mathcal{F}(X) \rightarrow \mathcal{G}(X)$ for all X in $\mathbf{C}$ such that for all $\alpha : X \rightarrow Y$ in $\mathbf{C}$, the diagram

$$\begin{array}{ccc} \mathcal{F}(X) & \xrightarrow{\mathcal{F}(\alpha)} & \mathcal{F}(Y) \\ \nu_X \downarrow & & \downarrow \nu_Y \\ \mathcal{G}(X) & \xrightarrow{\mathcal{G}(\alpha)} & \mathcal{G}(Y) \end{array}$$

commutes. A *natural isomorphism* is a natural transformation ν such that every ν_X is an isomorphism.

Natural transformations arise quite often. Whenever a morphism is referred to as *canonical* or *natural*, there is likely a natural transformation involved. The technical meaning of natural thus matches its use. A basic, but important, example is that of an adjoint functor.

Definition 1.7: Adjoint Functors

Let $\mathcal{C}, \mathcal{D}$ be categories, and let $\mathcal{F} : \mathcal{C} \rightarrow \mathcal{D}$, $\mathcal{G} : \mathcal{D} \rightarrow \mathcal{C}$ be covariant functors. We say $\mathcal{F}$ is left-adjoint to $\mathcal{G}$, and likewise $\mathcal{G}$ is right-adjoint to $\mathcal{F}$, if for all X in $\mathcal{C}$ and Y in $\mathcal{D}$, there exist natural isomorphisms

$$\mathrm{Hom}_{\mathcal{C}}(X, \mathcal{G}(Y)) \rightarrow \mathrm{Hom}_{\mathcal{D}}(\mathcal{F}(X), Y).$$

We say a functor $\mathcal{F}$ is a left-adjoint functor if it has a right adjoint, and vice versa.

For example, the free group on a given set is simply a group such that a set function from a set A to a group G is the same as a group homomorphism $F(A) \rightarrow G$. What this really means is that for all sets A and groups G , there are natural identifications

$$\mathrm{Hom}_{\mathrm{Set}}(A, S(G)) \cong \mathrm{Hom}_{\mathrm{Grp}}(F(A), G),$$

where $S : \mathrm{Grp} \rightarrow \mathrm{Set}$ is the forgetful functor. That means the functor $F : \mathrm{Set} \rightarrow \mathrm{Grp}$ which constructs free groups is left-adjoint to the forgetful functor S . In fact, this applies to any free construction we have in any category. As a rule, we say that the free functor is the left-adjoint to the forgetful functor.

We can thus find interesting functors as adjoints to more simple looking functors. Knowing that a functor is left-adjoint or right-adjoint simplifies many proofs, so this is a remarkable fact. Here is one way in which this happens.

Lemma 1.1

Right-adjoint functors commute with limits. That is, for $\mathcal{G} : \mathcal{D} \rightarrow \mathcal{C}$ a functor with a left-adjoint $\mathcal{F} : \mathcal{C} \rightarrow \mathcal{D}$, and $\mathcal{A} : \mathcal{I} \rightarrow \mathcal{D}$ a third functor, there exists a canonical isomorphism

$$\mathcal{G}(\varprojlim \mathcal{A}) \rightarrow \varprojlim (\mathcal{G} \circ \mathcal{A}).$$

Proof. Let $\mathcal{G} : \mathcal{D} \rightarrow \mathcal{C}$ be right adjoint to $\mathcal{F} : \mathcal{C} \rightarrow \mathcal{D}$, and let $\mathcal{A} : \mathcal{I} \rightarrow \mathcal{D}$ be a functor. Then recall that the limit is final with respect to cones

$$\begin{array}{ccc} & \varprojlim \mathcal{A} & \\ \lambda_I \swarrow & & \searrow \lambda_J \\ \mathcal{A}(I) & \xrightarrow{\alpha} & \mathcal{A}(J) \end{array}$$

with evident notation. Then $\mathcal{G}$ induces commutative diagrams in $\mathcal{C}$:

$$\begin{array}{ccc} & \mathcal{G}(\varprojlim \mathcal{A}) & \\ \mathcal{G}(\lambda_I) \swarrow & & \searrow \mathcal{G}(\lambda_J) \\ (\mathcal{G} \circ \mathcal{A})(I) & \xrightarrow{\mathcal{G}(\alpha)} & (\mathcal{G} \circ \mathcal{A})(J) \end{array}$$

By the universal property of limits, we have that

$$\begin{array}{ccc}
 & \mathcal{G}(\varprojlim \mathcal{A}) & \\
 & \downarrow \text{dashed} & \\
 & \varprojlim (\mathcal{G} \circ \mathcal{A}) & \\
 \swarrow & & \searrow \\
 (\mathcal{G} \circ \mathcal{A})(I) & \xrightarrow{\quad} & (\mathcal{G} \circ \mathcal{A})(J)
 \end{array}$$

commutes. Note here that there exists a unique morphism

$$\mathcal{G}(\varprojlim \mathcal{A}) \rightarrow \varprojlim (\mathcal{G} \circ \mathcal{A}).$$

By adjunction, the morphisms $\varprojlim (\mathcal{G} \circ \mathcal{A}) \rightarrow \mathcal{G} \circ \mathcal{A}(I)$ determine morphisms in $\mathbf{D}$ by the map

$$\mathrm{Hom}_{\mathbf{C}}(X, \mathcal{G}(Y)) \rightarrow \mathrm{Hom}_{\mathbf{D}}(\mathcal{F}(X), Y).$$

That is, each morphism determines a morphism

$$\mathcal{F}(\varprojlim (\mathcal{G} \circ \mathcal{A})) \rightarrow \mathcal{A}(I).$$

We then have the diagram in $\mathbf{C}$:

$$\begin{array}{ccc}
 & \mathcal{F}(\varprojlim (\mathcal{G} \circ \mathcal{A})) & \\
 \swarrow & & \searrow \\
 \mathcal{A}(I) & \xrightarrow{\quad} & \mathcal{A}(J)
 \end{array}$$

By the universal property defining $\varprojlim \mathcal{A}$,

$$\begin{array}{ccc}
 & \mathcal{F}(\varprojlim (\mathcal{G} \circ \mathcal{A})) & \\
 & \downarrow \text{dashed} & \\
 & \varprojlim \mathcal{A} & \\
 \swarrow & & \searrow \\
 \mathcal{A}(I) & \xrightarrow{\quad} & \mathcal{A}(J)
 \end{array}$$

By the adjunction condition, we have

$$\mathrm{Hom}_{\mathbf{D}}(\mathcal{F}(\varprojlim (\mathcal{G} \circ \mathcal{A})), \varprojlim \mathcal{A}) \cong \mathrm{Hom}_{\mathbf{C}}(\varprojlim (\mathcal{G} \circ \mathcal{A}), \mathcal{G}(\varprojlim \mathcal{A})).$$

Therefore, this morphism not only determines a morphism $\varprojlim (\mathcal{G} \circ \mathcal{A}) \rightarrow \mathcal{G}(\varprojlim \mathcal{A})$, but this morphism

remains unique. We thus have the diagram

$$\begin{array}{ccc}
 & (\varprojlim (\mathcal{G} \circ \mathcal{A})) & \\
 & \downarrow \text{dashed} & \\
 & \mathcal{G}(\varprojlim \mathcal{A}) & \\
 \swarrow & & \searrow \\
 \mathcal{G}(\mathcal{A}(I)) & \longrightarrow & \mathcal{G}(\mathcal{A}(J))
 \end{array}$$

Therefore, we have unique morphisms

$$\phi : \varprojlim (\mathcal{G} \circ \mathcal{A}) \rightarrow \mathcal{G}(\varprojlim \mathcal{A}), \quad \psi : \mathcal{G}(\varprojlim \mathcal{A}) \rightarrow \varprojlim (\mathcal{G} \circ \mathcal{A}).$$

Since these are unique, it follows fairly immediately that they are isomorphisms. We have

$$\phi\psi\phi = \phi, \quad \psi\phi\psi$$

by uniqueness, and therefore

$$\psi\phi = 1_{\mathcal{G}(\varprojlim \mathcal{A})}, \quad \phi\psi = 1_{\varprojlim (\mathcal{G} \circ \mathcal{A})}.$$

■

The book claims this outline does not verify that the needed diagrams commute, but I feel such verification is immediate from the fact that $\mathcal{F}, \mathcal{G}, \mathcal{A}$ are covariant functors and from the adjunction condition.

The dual of this lemma, which is also true, is that left-adjoint functors commute with colimits:

$$\mathcal{F}(\varinjlim \mathcal{A}) = \varinjlim (\mathcal{F} \circ \mathcal{A}).$$

This statement is fairly important, because it implies that right-adjoint functors preserve products and kernels and other constructions described by limits, provided they exist.

Exact functors are functors that preserve exactness. Since we have only studied exactness in $R\text{-Mod}$, let R, S be rings, and let $\mathcal{F} : R\text{-Mod} \rightarrow S\text{-Mod}$ be an additive functor; that is, the functions $\text{Hom}_R(A, B) \rightarrow \text{Hom}_S(\mathcal{F}(A), \mathcal{F}(B))$ are abelian group homomorphisms. Let

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightarrow{\psi} C \longrightarrow 0$$

be exact. Then $\mathcal{F}$ is exact if and only if

$$0 \longrightarrow \mathcal{F}(A) \xrightarrow{\mathcal{F}(\phi)} \mathcal{F}(B) \xrightarrow{\mathcal{F}(\psi)} \mathcal{F}(C) \longrightarrow 0$$

is exact (for all exact sequences).

Functors are not always exact, but it can often be more interesting when they are not exact. An additive functor is called *left-exact* if whenever

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightarrow{\psi} C$$

is exact, so is the induced complex

$$0 \longrightarrow \mathcal{F}(A) \xrightarrow{\mathcal{F}(\phi)} \mathcal{F}(B) \xrightarrow{\mathcal{F}(\psi)} \mathcal{F}(C)$$

Left-exact functors give rise to what are called *derived functors*, which are studied in homological algebra, and will be introduced in the next chapter.

It is fairly easily shown that right-adjoint additive functors are left-exact, since they preserve kernels. The dual statements hold as well; left-adjoint additive functors $\mathcal{G}$ are right-exact, meaning whenever

$$A \xrightarrow{\phi} B \xrightarrow{\psi} C \longrightarrow 0$$

is exact, then so is

$$\mathcal{G}(A) \xrightarrow{\mathcal{G}(\phi)} \mathcal{G}(B) \xrightarrow{\mathcal{G}(\psi)} \mathcal{G}(C) \longrightarrow 0$$

1.2 Tensor Products and the Tor Functor

For the rest of the chapter, R will denote a commutative ring, and we will work exclusively in $R\text{-Mod}$.

1.2.1 Bilinear Maps and the Definition of Tensor Product

We have seen that $M \oplus N$ serves as both the product and coproduct of R -modules M, N , and thus an R -module homomorphism $M \oplus N \rightarrow P$ is determined by homomorphisms $M \rightarrow P$ and $N \rightarrow P$. However, there is another way to map the set $M \times N$ into P in a way compatible with the R -module structure.

Definition 1.8: Bilinear

Let M, N, P be R -modules. A function $\varphi : M \times N \rightarrow P$ is R -bilinear if and only if

- For all $m \in M$, the map $n \mapsto \varphi(m, n)$ is an R -module homomorphism $N \rightarrow P$;
- for all $n \in N$, the map $m \mapsto \varphi(m, n)$ is an R -module homomorphism $M \rightarrow P$.

That is, if φ is R -bilinear, then it is linear in both components. However, note that φ itself is *not* linear, even when viewed as $M \oplus N \rightarrow P$. However, we would expect there is a way to treat φ as if it was R -linear. It is then natural to consider, using universal properties, a new R -module $M \otimes_R N$ together with an R -bilinear map

$$\otimes : M \times N \rightarrow M \otimes_R N,$$

such that *every* R -bilinear map factors uniquely through $M \otimes_R N$:

$$\begin{array}{ccc} M \times N & \xrightarrow{\varphi} & P \\ \otimes \downarrow & \nearrow \bar{\varphi} & \\ M \otimes_R N & & \end{array}$$

such that $\bar{\varphi}$ is an R -module homomorphism. This is the tensor product.

Definition 1.9: Tensor Product

The tensor product over R of two R -modules M, N , denoted $M \otimes_R N$, is an R -module such that every R -bilinear map factors uniquely through it, in a way that the induced map is an R -linear map.

The subscript R is important, since S -bilinearity may be different from R -bilinearity, if $R \neq S$. It is not unusual to drop the subscript when the ring is understood, but we will generally not do this.

Lemma 1.2

Tensor product exist in $R\text{-Mod}$.

Proof. Let M, N be R -modules. Recall that the free R -module $F^R(M \times N)$ comes with a set map

$$j : M \times N \rightarrow F^R(M \times N)$$

which is universal with respect to functions $M \times N \rightarrow P$ for any R -module P . What we need is to make this into an R -bilinear map. This means we need to identify elements $j(m, n_1 + n_2)$ with $j(m, n_1) + j(m, n_2)$, and similarly with m . Let K be the free R -submodule generated by all elements of the form

$$j(m, r_1 n_1 + r_2 n_2) - r_1 j(m, n_1) - r_2 j(m, n_2)$$

and

$$j(r_1 m_1 + r_2 m_2, n) - r_1 j(m_1, n) - r_2 j(m_2, n).$$

Define

$$M \otimes_R N := \frac{F^R(M \times N)}{K},$$

and let $\otimes$ be the map πj , where π is the natural projection:

$$\otimes : M \times N \xrightarrow{j} F^R(M \times N) \xrightarrow{\pi} \frac{F^R(M \times N)}{K} =: M \otimes_R N$$

It is fairly obvious that this is an R -bilinear map at this point, since in the quotient module we will have

$$j(m, r_1 n_1 + r_2 n_2) - r_1 j(m, n_1) - r_2 j(m, n_2) = 0 \implies j(m, r_1 n_1 + r_2 n_2) = r_1 j(m, n_1) + r_2 j(m, n_2),$$

and similarly for m . We now show that this construction satisfies the universal property. Let $\phi : M \times N \rightarrow P$ be any R -bilinear map. The universal property of free modules induces a homomorphism $\tilde{\phi}$ such that

$$\begin{array}{ccc} M \times N & \xrightarrow{\phi} & P \\ \downarrow j & \nearrow \tilde{\phi} & \\ F^R(M \times N) & & \end{array}$$

commutes. Note that $\tilde{\phi}$ will factor uniquely through the quotient by K if $\ker \tilde{\phi} = K$. By commutativity of the diagram, this is immediate:

$$\begin{aligned} & \tilde{\phi}(j(m, r_1 n_1 + r_2 n_2) - r_1 j(m, n_1) - r_2 j(m, n_2)) \\ &= \tilde{\phi}(j(m, r_1 n_1 + r_2 n_2) - \tilde{\phi}(r_1 j(m, n_1)) - \tilde{\phi}(r_2 j(m, n_2))) \\ &= \phi(m, r_1 n_1 + r_2 n_2) - r_1 \phi(m, n_1) - r_2 \phi(m, n_2) \\ &= 0. \end{aligned}$$

The last step follows because ϕ is R -bilinear. It is easily shown that the converse is true; that if $\tilde{\phi}(x) = 0$,

then x is of this form. Therefore, $\tilde{\phi}$ factors uniquely through the quotient by K :

$$\begin{array}{ccc}
 M \times N & \xrightarrow{\phi} & P \\
 \downarrow j & \nearrow \tilde{\phi} & \uparrow \\
 F^R(M \times N) & & \\
 \downarrow \pi & \nearrow \bar{\phi} & \\
 M \otimes_R N & &
 \end{array}$$

$\otimes$ (curved arrow from $M \times N$ to $M \otimes_R N$)

By commutativity of the diagram, we are done. ■

One useful fact from this explicit construction is that elements of $M \otimes_R N$ arise from finite R -linear combinations

$$\sum_i r_i (m_i \otimes n_i).$$

The R -bilinearity of the map $\otimes$ amounts to the statements that

$$m \otimes (n_1 + n_2) = (m \otimes n_1) + (m \otimes n_2),$$

$$(m_1 + m_2) \otimes n = (m_1 \otimes n) + (m_2 \otimes n),$$

$$m \otimes (rn) = (rm) \otimes n = r(m \otimes n).$$

It follows that elements of a tensor product can be viewed as

$$\sum_i r_i (m_i \otimes n_i) = \sum_i (r_i m_i) \otimes n_i.$$

Tensors that require only one term in the above summation; that is, tensors of the form $m \otimes n$, are called *pure tensors*. Not all tensors are pure tensors, but pure tensors turn out to be somewhat useful, since they can generate the tensor product. For example, morphisms $\alpha, \beta : M \otimes_R N \rightarrow P$ are the same if they coincide on all pure tensors.

1.2.2 Adjunction with Hom and Explicit Computations

Using the tensor product, we can define a collection of new covariant functors. Let N be any R -module. Then the map $M \mapsto M \otimes_R N$ is a covariant functor $R\text{-Mod} \rightarrow R\text{-Mod}$. The action on morphisms is given as follows: for any morphism $\alpha : M_1 \rightarrow M_2$, there is an induced R -bilinear map $\alpha \times \text{id}_N$, and thus an induced R -linear map

$$\alpha \otimes N : M_1 \otimes_R N \rightarrow M_2 \otimes_R N.$$

This map is thus defined on pure tensors as $m \otimes n \mapsto \alpha(m) \otimes n$. Functoriality follows immediately, and we have a covariant functor.

Recall that our definition of bilinear means that any R -bilinear map $\phi : M \times N \rightarrow P$ determines R -linear maps $\phi(-, n)$ and $\phi(m, -)$. That is, ϕ determines a map

$$M \rightarrow \text{Hom}_R(N, P).$$

Moreover, this map is an R -linear homomorphism (recall that hom-sets are R -modules with scalar multiplication given by $(r\phi)(x) = r\phi(x)$). That is, this bilinear map can be viewed as an element of

$$\text{Hom}_R(M, \text{Hom}_R(N, P)).$$

These considerations make the following statements fairly natural.

Lemma 1.3

For all R -modules M, N, P , there is an isomorphism of R -modules

$$\mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P)) \cong \mathrm{Hom}_R(M \otimes_R N, P).$$

Proof. Every $\alpha \in \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$ determines an R -bilinear map by

$$\phi : M \times N \rightarrow P, \quad (m, n) \mapsto \alpha(m)(n).$$

By the universal property of tensors, ϕ factors uniquely through $\otimes$ as an R -linear map $\bar{\phi} : M \otimes_R N \rightarrow P$. It will be shown in the exercises that $\alpha \mapsto \bar{\phi}$ is linear, and an inverse will be constructed. ■

Corollary 1.1

For every R -module N , the functor $- \otimes_R N$ is left-adjoint to the functor $\mathrm{Hom}_R(N, -)$.

Proof. By definition 1.7, these functors are adjoint if and only if

$$\mathrm{Hom}_R(- \otimes_R N, -) \cong \mathrm{Hom}_R(-, \mathrm{Hom}_R(N, -)).$$

That is, for all R -modules M, P , we have

$$\mathrm{Hom}_R(M \otimes_R N, P) \cong \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P)).$$

This is precisely the result of lemma 1.3. ■

It follows that the functor $- \otimes_R N$ preserves colimits, and by commutativity of the tensor product we have that $M \otimes_R -$ does as well. In particular, these functors are right-exact functors. These observations have many important consequences.

Corollary 1.2

For all R -modules M, N, P ,

$$(M \oplus N) \otimes_R P \cong (M \otimes_R P) \oplus (N \otimes_R P).$$

Proof. Since $(M \oplus N)$ is the colimit of some functor, and since $- \otimes_R P$ is left-adjoint by corollary 1.1, it will commute with coproducts by the dual of lemma 1.1. In fact, it will commute for (possibly infinite) direct sums:

$$\left(\bigoplus_{\alpha \in A} M_\alpha \right) \otimes_R N \cong \bigoplus_{\alpha \in A} (M_\alpha \otimes_R N).$$

This statement computes all tensors for tensor products of free R -modules. ■

Corollary 1.3

For any two sets A, B ,

$$R^{\oplus A} \otimes_R R^{\oplus B} \cong R^{\oplus A \times B}.$$

Proof. By the previous corollary,

$$R^{\oplus A} \otimes_R R^{\oplus B} \cong (R^{\oplus A})^{\oplus B}.$$

This has been shown previously to be isomorphic to $R^{\oplus A \times B}$. ■

This shows that bases and generating sets work as expected - if $\{e_i\}$ and $\{f_i\}$ generate M, N respectively, then the pure tensors $\{e_i \otimes f_i\}$ generates $M \otimes_R N$. If the modules are free and these sets are minimal, then this is a basis for the tensor product. This is all that can really happen over a field - for finitely generated spaces of dimension n and m , the tensor product just yields a space of dimension mn . Tensor products are thus more interesting over general rings.

Corollary 1.4

For all R -modules N and ideals I of R ,

$$\frac{R}{I} \otimes_R N \cong \frac{N}{IN}.$$

Proof. Since $- \otimes_R N$ is right-exact, the sequence

$$0 \longrightarrow I \longrightarrow R \longrightarrow \frac{R}{I} \longrightarrow 0$$

induces the exact sequence

$$I \otimes_R N \longrightarrow R \otimes_R N \longrightarrow \frac{R}{I} \otimes_R N \longrightarrow 0$$

Note that $R \otimes_R N \cong N$. The image of $I \otimes_R N$ in this R -module is generated by the image of pure tensors $a \otimes n$ for $a \in I, n \in N$. This is IN because I is an ideal, and since the sequence is exact, the kernel of the R -module homomorphism $N \rightarrow \frac{R}{I} \otimes_R N$ is IN . This completes the proof by the first isomorphism theorem. ■

Corollary 1.5

For all ideals I, J of R ,

$$\frac{R}{I} \otimes_R \frac{R}{J} \cong \frac{R}{I+J}.$$

Proof. This follows immediately from the previous corollary and the third isomorphism theorem. ■

1.2.3 Exactness Properties of Tensor; Flatness

Although the tensor product is left-adjoint and thus right-exact, it is in general not an exact functor. For example, in the proof of corollary 1.4, the map

$$I \otimes_R N \rightarrow N$$

induced by the inclusion $I \hookrightarrow R$ may not be injective, and thus a short sequence would fail to be exact at N .

For another example, consider the injection $\mathbb{Z} \rightarrow \mathbb{Z}$ given by multiplication by 2. Tensoring by $\mathbb{Z}/2\mathbb{Z}$

over $\mathbb{Z}$, and keeping in mind that $R \otimes_R N \cong N$, we have the map (still given by multiplication by 2)

$$\frac{\mathbb{Z}}{2\mathbb{Z}} \rightarrow \frac{\mathbb{Z}}{2\mathbb{Z}}.$$

Note that this map sends $[0], [1] \mapsto 0$. This is thus the zero morphism, and clearly the exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z} \longrightarrow \frac{\mathbb{Z}}{2\mathbb{Z}} \longrightarrow 0$$

is no longer exact after tensoring.

However, if N is free, then $- \otimes_R N$ is exact. For every inclusion $M_1 \hookrightarrow M_2$, tensoring by $N = R^{\oplus A}$ gives

$$M_1^{\oplus A} \hookrightarrow M_2^{\oplus A}$$

which is given once again by the inclusion $M_1^{\oplus A} \subseteq M_2^{\oplus A}$. Due to this fact, tensoring is exact in $k\text{-Vect}$. That is, if

$$0 \longrightarrow V_1 \longrightarrow V_2 \longrightarrow V_3 \longrightarrow 0$$

is exact in $k\text{-Vect}$, then for all k -vector spaces W ,

$$0 \longrightarrow V_1 \otimes_k W \longrightarrow V_2 \otimes_k W \longrightarrow V_3 \otimes_k W \longrightarrow 0$$

is exact. More simply, if $V_\bullet$ is exact, then $V_\bullet \otimes_k W$ is exact.

Definition 1.10: Flat

An R -module N is flat if the functor $_- \otimes_R N$ is exact.

Flat modules are very important, especially in areas like algebraic geometry. It then makes sense to study the exactness of R -modules.

1.2.4 The Tor Functors

Now we will introduce the Tor functors, which are an important class of what is called a derived functor. The failure of the functor $- \otimes_R N$ to be exact is measured by the other functor $\text{Tor}_1^R(-, N)$. For example, if N is flat, then $\text{Tor}_1^R(M, N) = 0$ for all R -modules M . Moreover, if

$$0 \longrightarrow M \longrightarrow N \longrightarrow P \longrightarrow 0$$

is exact, then tensoring by any A gives a new exact sequence

$$\text{Tor}_1^R(P, A) \longrightarrow M \otimes_R A \longrightarrow N \otimes_R A \longrightarrow P \otimes_R A \longrightarrow 0$$

If the module on the left vanishes, then every exact sequence ending in C remains exact under $- \otimes_R N$ in this case. This process can in fact be continued to obtain a longer exact complex

$$\text{Tor}_1^R(A, N) \longrightarrow \text{Tor}_1^R(B, N) \longrightarrow \text{Tor}_1^R(C, N) \longrightarrow A \otimes_R N \longrightarrow B \otimes_R N \longrightarrow C \otimes_R N \longrightarrow 0$$

There in fact exist even more functors $\text{Tor}_2^R(-, N)$, $\text{Tor}_3^R(-, N)$ that continue this complex further. These are called the *derived functors* of tensor.

This is how we compute these functors. Recall the definition of a free resolution of a module M : a free resolution of M is an *exact* complex $M_\bullet$ of free R -modules

$$\dots \longrightarrow R^{\oplus S_2} \longrightarrow R^{\oplus S_1} \longrightarrow R^{\oplus S_0} \longrightarrow M \longrightarrow 0.$$

If we throw away M and tensor the free part by N , we get the complex $M_\bullet \otimes_R N$

$$\dots \longrightarrow N^{\oplus S_2} \longrightarrow N^{\oplus S_1} \longrightarrow N^{\oplus S_0} \longrightarrow 0.$$

This follows from the fact that tensor commutes with colimits. The homology of this complex will suprisingly not depend (up to isomorphism) on the chosen free resolution, so we can define

$$\mathrm{Tor}_i^R(M, N) := H_i(M_\bullet \otimes_R N).$$

By this definition, we will show in the exercises that $\mathrm{Tor}_0^R(M, N) = M \otimes_R N$, and if N is flat, then $\mathrm{Tor}_i^R(M, N) = 0$ for all $i > 0$ (because tensoring by N would be exact, so it would return an exact sequence, and would thus have no homology). In fact, this shows that if $\mathrm{Tor}_1^R(M, N) = 0$ for all M , then $\mathrm{Tor}_i^R(M, N) = 0$ for all i and for all M . Then N is flat.

We will later formalize these notions further, but for now we will simply assume definitions can be made as such. For now, we will do some simple computation with these functors. We have shown that every PID has a free resolution of length 1

$$0 \longrightarrow R^{\oplus m_1} \longrightarrow R^{\oplus m_0} \longrightarrow M \longrightarrow 0$$

This property characterizes PIDs. Moreover, if

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

is exact, then there exists compatible resolutions

$$\begin{array}{ccccccc} & & 0 & & 0 & & 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & R^{\oplus a_1} & \longrightarrow & R^{\oplus b_1} & \longrightarrow & R^{\oplus c_1} \longrightarrow 0 \\ & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma \\ 0 & \longrightarrow & R^{\oplus a_0} & \longrightarrow & R^{\oplus b_0} & \longrightarrow & R^{\oplus c_0} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 0 & & 0 & & 0 \end{array}$$

such that columns and rows are exact. Tensoring the free rows by N gives the exact diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & N^{\oplus a_1} & \longrightarrow & N^{\oplus b_1} & \longrightarrow & N^{\oplus c_1} \longrightarrow 0 \\ & & \downarrow \alpha \otimes N & & \downarrow \beta \otimes N & & \downarrow \gamma \otimes N \\ 0 & \longrightarrow & N^{\oplus a_0} & \longrightarrow & N^{\oplus b_0} & \longrightarrow & N^{\oplus c_0} \longrightarrow 0 \end{array}$$

Applying the snake lemma gives the exact sequence

$$\begin{array}{ccccccc} 0 & \longrightarrow & H_1(A_\bullet \otimes_R N) & \longrightarrow & H_1(B_\bullet \otimes_R N) & \longrightarrow & H_1(C_\bullet \otimes_R N) \\ & & & & \searrow \delta & & \\ H_0(A \otimes_R N) & \longrightarrow & H_0(B \otimes_R N) & \longrightarrow & H_0(C \otimes_R N) & \longrightarrow & 0 \end{array}$$

This is precisely the sequence of Tor modules

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \mathrm{Tor}_1^R(A, N) & \longrightarrow & \mathrm{Tor}_1^R(B, N) & \longrightarrow & \mathrm{Tor}_1^R(C, N) \\
 & & & & \searrow \delta & & \\
 \mathrm{Tor}_0^R(A, N) & \longrightarrow & \mathrm{Tor}_0^R(B, N) & \longrightarrow & \mathrm{Tor}_0^R(C, N) & \longrightarrow & 0
 \end{array}$$

The 0 on the left comes from the fact that Tor_2^R vanishes if R is a PID. Moreover, Tor_i^k vanishes for all $i > 0$ if k is a field, since vector spaces are flat.